

Spaces Product Decision Framework	
Question	Result
Does it solve a real problem?	✅ / ❌
Which Space templates benefit?	Family / Business / Community / Friends
Is it better as a new module or an existing module?	Existing / New
Does it fit the vision of Spaces?	✅ / ⚠️ / ❌
Complexity vs. value	Low / Medium / High
Recommendation	Build / Defer / Reject

Spaces v2

Roles & Permissions Specification (Final)

Philosophy

Spaces has five fixed roles:

- * Owner
- * Admin
- * Moderator
- * Member
- * Guest

These role names are permanent and cannot be renamed or deleted.

Each role has a predefined default permission set.

When a member is assigned a role, they automatically receive that role's default permissions.

The Space Owner may grant or revoke individual permissions for a specific member without changing the role itself.

This allows exceptions for individual members while keeping the role system simple and consistent.

Default Roles

Permission	Owner	Admin	Moderator	Member	Guest
Manage Space Settings	✓	✗	✗	✗	✗
Manage Members	✓	✓	✗	✗	✗
Manage Roles	✓	✗	✗	✗	✗
Manage Modules	✓	✓	✗	✗	✗

Permission	Owner	Admin	Moderator	Member	Guest
Invite Members	✓	✓	✗	✗	✗
Remove Members	✓	✓	✗	✗	✗
Post Pings	✓	✓	✓	✓	✗
Create Events	✓	✓	✓	✓	✗
Upload Photos/ Videos	✓	✓	✓	✓	✗
Upload Files	✓	✓	✓	✓	✗
Edit Own Content	✓	✓	✓	✓	✗
Delete Own Content	✓	✓	✓	✓	✗
Delete Others' Content	✓	✓	✗	✗	✗
View Content	✓	✓	✓	✓	✓

Role Rules

Owner

- * There is always one Owner.
- * Full control of the Space.
- * Can transfer ownership.
- * Can delete the Space.
- * Can assign roles.
- * Can grant or revoke individual permissions for any member.

- * Cannot be removed without first transferring ownership.

Admin

- * Assists the Owner in managing the Space.
- * Can manage members.
- * Can manage modules.
- * Can invite and remove members.
- * Cannot delete the Space.
- * Cannot transfer ownership.
- * Cannot manage roles.
- * Cannot remove the Owner.

Moderator

- * Trusted participant.
- * Can fully participate in the Space.
- * Can create and manage their own content.
- * Cannot manage members.
- * Cannot invite members.
- * Cannot manage modules.
- * Cannot delete content created by others.

Member

- * Standard participant.
- * Can participate in all normal Space activities.
- * Can only manage their own content.
- * Has no administrative permissions.

Guest

- * Read-only by default.
- * Can only view content.
- * Cannot post, upload, edit, or create anything unless explicitly granted through an individual permission override.

Permission Override System

Roles determine the default permissions.

The Owner may override permissions for an individual member only.

Overrides do not modify the role itself.

Example:

Sarah

Role:
Member

Default Permissions:

- ✓ Post Pings
- ✓ Create Events
- ✓ Upload Photos/Videos
- ✓ Upload Files

Owner Override:
+ Invite Members

Final Permissions:

- ✓ Post Pings
- ✓ Create Events
- ✓ Upload Photos/Videos
- ✓ Upload Files
- ✓ Invite Members

Another Member assigned the same role would still have the normal Member permissions.

Permission Resolution

Permissions are determined in the following order:

1. Assigned Role
2. Load the role's default permissions.
3. Apply any user-specific permission overrides.
4. The resulting permission set becomes the member's effective permissions.

Design Goals

- * Five consistent, easy-to-understand roles.
- * Predictable default permissions.
- * Flexible per-user exceptions.
- * No custom role creation.
- * No editing role definitions.
- * Individual overrides without affecting other members.

Module Management Specification (Final)

Philosophy

Modules are permanent features of Spaces.

A module is never created, deleted, installed, or uninstalled.

Instead, each Space controls whether an available module is enabled or disabled.

Disabling a module hides it from the Space without deleting any data.

Re-enabling the module restores it exactly as it was.

Required Modules

The following modules exist in every Space and cannot be disabled.

Module	Required	Can Disable
Space Pings	✓	✗
Photos	✓	✗
Members	✓	✗
Settings	✓	✗

Optional Modules

Optional modules may be enabled or disabled by members with the appropriate permissions.

Current optional modules:

- Polls
-
- Events
-
- Files

* Future modules will follow the same architecture.

Module States

Every module always exists in one of two states.

State	Description
Enabled	Visible and fully functional within the Space.
Disabled	Hidden from the Space while preserving all data and settings.

* There is no deleted state.

Disabling a Module

Disabling a module will:

- * Hide the module from the Space.
- * Hide all of the module's content.
- * Preserve all existing data.
- * Preserve all module settings.
- * Preserve permissions.
- * Preserve activity/history.

No information is deleted.

Re-enabling a Module

Re-enabling a module restores:

- * All previous content.
- * All previous settings.
- * All permissions.
- * All history.

The module returns exactly as it existed before being disabled.

Module Permissions

Only members with the appropriate permission (Owner/Admin by default) may:

- * Enable modules.

- * Disable modules.
- * Configure module settings.

Other members can only use modules that are enabled for the Space.

Module Independence

Each module is responsible for its own:

- * Content
- * Data
- * Settings
- * User interface
- * Functionality

Modules do not depend on other optional modules in order to function.

Design Goals

- * Permanent module architecture.
- * No module deletion.
- * Simple enable/disable management.
- * Safe data preservation.
- * Consistent behavior across all modules.
- * Scalable foundation for future modules.
- * Required modules always available.
- * Optional modules configurable per Space.

Current Module List

Module	Type	Default State
Space Pings	Required	Enabled
Photos	Required	Enabled
Members	Required	Enabled
Settings	Required	Enabled
Polls	Optional	Template dependent
Events	Optional	Template dependent

Module	Type	Default State
Files	Optional	Template dependent

Space Templates & Default Modules

When creating a new Space, the selected Template Type determines which optional modules are enabled by default.

Templates do not create different kinds of modules.

Templates only define the starting module setup for that Space.

Template Types

Current template types:

- * Family
- * Friends
- * Business
- * Community
- * Custom

Template-Based Module Defaults

Each template has a preset group of enabled modules.

Template	Default Enabled Modules
Family	Space Pings, Photos, Events, Members
Business	Space Pings, Files, Events, Members
Custom	User-selected modules

Custom Template

Custom allows the creator to manually choose which modules are enabled when the Space is created.

Even in Custom, required modules still apply:

- * Space Pings
- * Photos
- * Members
- * Settings

Optional modules can be turned on or off during creation.

After Creation

After a Space is created:

- * Owners/Admins can enable optional modules.
- * Owners/Admins can disable optional modules.
- * Required modules stay enabled.
- * Disabled module data is hidden, not deleted.
- * Re-enabled modules restore existing content and settings.

E2EE Specification Final

Scope

Spaces v1 already has E2EE for:

- * Pings
- * Messages

Spaces v2 expands the same existing E2EE system to:

- * Photos
- * Videos
- * Files

There will not be a separate encryption system for media.

Core Rule

Photos, videos, and files must use the existing v1 Pings/messages E2EE architecture.

That means:

- * Same encryption model
- * Same Firebase UID-based identity model
- * Same member access behavior
- * Same overall E2EE process

Member Access

New Members

When a new member joins a Space:

- * They can access existing encrypted media.
- * They can access future encrypted media.
- * This applies to photos, videos, and files.

Removed Members

When a member is removed from a Space:

- * The Space disappears from their account.
- * They cannot access the Space anymore.
- * They cannot view, open, download, or decrypt Space media through the app anymore.

Saved Media

If a member saved media to their physical device before being removed, Spaces does not control that local copy.

Encrypted Content

These should be encrypted:

Item	Encrypt
Photo content	
Video content	
File content	
Captions	
Descriptions	

Unencrypted Technical Metadata

These may remain unencrypted because the app needs them to function:

Item	Encrypt
Upload timestamp	
Uploader ID	
Space ID	
Module ID	
File type	
File size	

Device / Account Model

Spaces v2 media encryption uses the existing v1 model:
Firebase UID = encryption identity

If the user signs in and receives the same Firebase UID, they can access their encrypted Spaces content according to membership and permissions.

There is no new trusted-device approval flow in v2.

Content Ownership

Once media is uploaded to a Space, it becomes part of that Space.

The uploader is recorded as the creator, but the content belongs to the Space.

If the uploader leaves or is removed:

- * Their uploaded photos remain.
- * Their uploaded videos remain.
- * Their uploaded files remain.
- * Remaining members keep access based on permissions.
- * The removed member loses app access to that Space and its encrypted media.

Deletion

Removing a member does not delete their uploaded media.

Media can only be deleted by members with the correct permissions.

Customization Specification (Final)

Philosophy

Spaces v2 allows users to customize the organization of both Spaces and modules without changing the functionality of the Space.

Customization only affects layout and navigation. It does not affect permissions, data, or module behavior.

Module Ordering

Module ordering controls the order in which modules appear within a Space.

Each Space maintains its own module order.

All members of a Space see the same module order.

Default Module Order

When a Space is created, the selected template determines:

- * Which optional modules are enabled.
- * The initial module order.

Owners and Admins may change the module order at any time.

Module Reordering

Members with the appropriate permissions (Owner/Admin by default) may reorder modules using drag and drop.

Modules can be moved to any position in the Space.

Required modules and optional modules may both be reordered.

Disabled Modules

Disabled modules:

- * Do not appear in the Space.
- * Do not affect the order of visible modules.

If a disabled module is re-enabled:

- * It returns to its previous position in the module order.
- * Its data and settings remain intact.

Module Ordering Permissions - Owner and admin only

Space Ordering

Space ordering controls the order in which Spaces appear within a user's Home screen/sidebar.

Unlike module ordering, Space ordering is personal.

Each user maintains their own Space order.

Changing the order of Spaces only affects that user's account.

No other member sees the change.

Space Reordering

Users may reorder their Spaces using drag and drop.

A user's preferred order is synchronized to their account.

New Spaces

When a user:

- * Creates a new Space, or
- * Joins an existing Space,

the new Space is placed at the bottom of that user's Space list by default.

The user may move it anywhere afterward.

Design Goals

- * Simple drag-and-drop organization.
- * Consistent module layout for all members of a Space.
- * Personalized Space organization for each user.
- * No impact on permissions or data.
- * Disabled modules preserve their position when re-enabled.
- * Customization affects presentation only and never alters module functionality or Space behavior.

Spaces v2.1

Module Interoperability Principle

Guiding Principle

Modules in Spaces should remain independent, but they should be able to reference each other when it improves context.

Each module owns its own data. Other modules may link to that data without duplicating it.

Core Rule

Modules are independent, but interconnected.

Examples

- * An Event can reference a List.
- * An Event can reference a Note.
- * A Note can reference Media.
- * A Note can reference Files.
- * A List can reference an Event.
- * A future Project can reference Boards, Files, Notes, Media, Reports, and Proposals.

Purpose

This prevents Spaces from feeling like disconnected tools. It allows each Space to become a connected system where related information can be linked together without merging modules or duplicating content.

Modules are independent but interconnected.

- * Modules own their own data.
- * Reference data instead of duplicating it.
- * A module should represent a complete workflow, not a single object.
- * If a feature naturally belongs in an existing module, don't create a new module.
- * Templates determine which modules are enabled—not what modules exist.
- * Required modules cannot be disabled; optional modules can be enabled or disabled by the Space Owner.

Module Category 1: Communication

1. Announcements

Purpose

A dedicated place for important Space-wide updates that shouldn't get buried in Space Pings.

Why?

Unlike Space Pings, announcements are intended to be:

- * Important
- * Persistent
- * Easy to find
- * Controlled by specific roles

Examples:

Family

- * Thanksgiving dinner at 3 PM
- * Vacation itinerary finalized
- * Grandpa's birthday party

Business

- * Office closed Monday
- * New company policy
- * Quarterly meeting

Community

- * Event postponed
- * Volunteer schedule
- * Membership renewal

Permissions

By default:

Role	Create	Edit Own	Delete Own	Delete Others
Owner	✓	✓	✓	✓
Admin	✓	✓	✓	✓
Moderator	✗	✗	✗	✗
Member	✗	✗	✗	✗
Guest	✗	✗	✗	✗

As with everything else in Spaces, these can be overridden through individual permissions if the Owner chooses.

Core Features

- * Rich text
- * Images
- * Videos
- * Files
- * Links
- * Pin announcements
- * Expiration date (optional)
- * Comments (optional, Owner controlled)
- * Reactions
- * Cross-module links

Cross-Module Linking

Announcements can reference:

- * Events
- * Notes
- * Lists
- * Media
- * Files

Example:

 Company Picnic

 Event
2027 Company Picnic

 Notes
Parking Instructions

 Media
Venue Photos

Comments

Owner-controlled (per announcement) 

When creating an announcement, the author can choose:

- * Comments Enabled
- * Comments Disabled

This keeps announcements flexible.

Examples:

- *  “Office closes at 5 PM.” → Comments Off
 - *  “Vote on the new logo.” → Comments On
-

Reactions

Enabled 

Members can react to announcements.

Examples:

- * 
- * 
- * 
- * 
- * 

This gives lightweight feedback without requiring comments.

Read Receipts

Not included ❌

No “Seen by 18 members.”

I agree this adds complexity without enough value for Spaces.

2. Rooms (Discussions)

Purpose

Rooms allow conversations inside a Space to be organized by topic, project, interest, or purpose instead of everything happening in one Space Ping feed.

Unlike direct messages, Rooms belong to the Space and are visible only to members with access.

Why?

As a Space grows, one conversation becomes difficult to follow.

Rooms keep conversations organized.

Family

- General
- Vacation Planning
- Christmas
- House Remodel

Friends

- Weekend Plans
- Gaming
- Fantasy Football
- Memes

Business

- General
- Marketing
- Development
- Leadership

Community

- General
- Volunteers
- Events
- Board Members

Permissions

Role	Create	Edit	Delete	Manage Members
Owner	✓	✓	✓	✓
Admin	✓	✓	✓	✓
Moderator	✓	✓ (Own)	✓ (Own)	✗
Member	✗	✗	✗	✗
Guest	✗	✗	✗	✗

All permissions remain customizable through the Roles & Permissions system.

Core Features

- Dedicated conversation feed
- Media sharing
- File sharing
- Reactions
- Replies
- Pinned messages
- Search
- End-to-End Encryption (E2EE)
- Cross-module linking

Cross-Module Linking

Rooms can reference:

Current Modules

- Events
- Lists
- Notes
- Media
- Files
- Polls
- Announcements

Future Modules

- Projects
- Boards
- Bids
- Reports
- Library

Example

Camping Trip Room

Linked Modules:

- Event → Camping Weekend
- Notes → Packing Notes
- List → Camping Checklist
- Media → Campsite Photos

Everything stays together around that topic instead of being scattered throughout Space Pings.

1. Public & Private Rooms

Rooms can be either:

- Public (Visible to everyone in the Space)
- Private (Visible only to selected members)

Private Rooms are intended primarily for Business and Community Spaces but are available in any Space.

Examples:

Family

- General
- Vacation
- Parents 

Business

- General
- Marketing
- Leadership 
- HR 

Community

- General
- Volunteers
- Board Members 

2. Room Organization

Rooms will remain a single flat list.

No categories or folders in v2.1.

This keeps navigation simple and leaves room for future enhancements if large Spaces eventually need additional organization.

3. Room Types

There is only one Room type.

Room behavior is controlled entirely through permissions.

Examples:

Read-only Room

- Only Owners/Admins can post.

Discussion Room

- Everyone with permission can post.

Private Room

- Only invited members can view and participate.

Future features (such as voice) will be capabilities of a Room rather than separate Room types.

Module Category 2: Content

Category 2: Content

v2.1 Change:

- Rename the existing Photos module to Media.
- Keep Files as its own separate module.
- The category name is Content.
- Content contains Media and Files.

Modules:

- Media
- Files

Module Category 3: Agenda

1. Lists

Purpose

Lists allow members of a Space to create shared checklists, supply lists, planning lists, and simple task-style lists without turning Spaces into a task management app.

Lists are designed for lightweight collaboration and organization across every type of Space.

Why?

Lists solve common shared planning needs across all Space templates.

Family

- Grocery Lists
- Packing Lists
- Christmas Gift Lists
- Home Repair Lists

Friends

- Trip Packing Lists
- BBQ Supply Lists
- Movie Watchlists
- Game Night Snacks

Business

- Office Supply Lists
- Launch Checklists
- Meeting Preparation
- Equipment Lists

Community

- Volunteer Supply Lists
- Event Setup Lists
- Donation Needs Lists
- Cleanup Checklists

Permissions

	Role	Create	Edit Own	Edit Any	Delete Own	Delete Any
	Owner	☒	☒	☒	☒	☒
	Admin	☒	☒	☒	☒	☒

Moderator					
Member					
Guest					

All permissions remain customizable through the Roles & Permissions system.

Core Features

- Create shared lists
- Add, edit, and delete list items
- Check and uncheck items
- Reorder list items
- Assign list items to members using @mentions
- Optional due dates on individual list items
- Optional notes/details for individual list items
- Attach Media and Files to a list
- Cross-module linking
- Search
- End-to-End Encryption (E2EE)

Cross-Module Linking

Lists can reference:

Current Modules

- Events
- Notes
- Rooms
- Announcements
- Media
- Files
- Polls

Future Modules

- Projects
- Boards
- Reports

Example

Camping Checklist

Linked Event

- Camping Weekend

Items

☒ Tent

Assigned: @Ian

☒ Sleeping Bags

Assigned: @Sarah

☐ Cooler

Assigned: @Ian

☐ Firewood

☐ First Aid Kit

Linked Notes

- Packing Notes

Linked Media

- Campsite Photos

Everything related to the camping trip remains connected without duplicating information.

Final Design Decisions

1. Assignable List Items

List items can be assigned to one or more members.

Assignments are made using @mentions.

Example:

☐ Bring Cooler

Assigned: @Ian

☐ Pick Up Ice

Assigned: @Sarah

Assignments are intended for accountability only and are ****not**** a replacement for Projects or Boards.

2. Due Dates

Due dates are optional for individual list items.

Examples:

☐ Buy Groceries

Due: Friday, 5:00 PM

☐ Reserve Campsite

Due: July 18

Lists themselves may also be linked to an Event when appropriate.

3. Comments

Lists do not support comments.

If discussion is needed, members should use:

- Space Pings
- Rooms
- Notes

This keeps Lists focused on lightweight organization instead of becoming a discussion or task management module.

Open Design Decision

List Sections

Should Lists support sections/groups?

Example:

Camping Trip

Food

- ☐ Hot Dogs
- ☐ Buns
- ☐ Chips

Gear

- ☐ Tent
- ☐ Lantern
- ☐ Chairs

Clothing

- ☐ Jacket
- ☐ Boots
- ☐ Socks

Or should every List remain a single continuous checklist?

This is the only remaining architectural decision before the Lists module is considered complete.

2. Notes

Purpose

Notes provide a centralized place to create, organize, and share long-form information within a Space.

Unlike Lists, which focus on action items, Notes focus on preserving knowledge, documentation, and reference information.

Notes serve as the primary documentation module for every Space.

Why?

Notes provide a permanent place for information that members may need to reference repeatedly.

Family

- Wi-Fi Password
- Emergency Contacts
- House Rules
- Vacation Itinerary
- Family Recipes

Friends

- Trip Plans
- Party Information
- Movie Recommendations
- Shared Ideas
- Gift Ideas

Business

- Meeting Notes
- Brainstorming
- Standard Operating Procedures
- Client Information
- Internal Documentation

Community

- Meeting Minutes
- Volunteer Guide
- Event Documentation

- Bylaws
- Policies

Permissions

Role	Create	Edit Own	Edit Any	Delete Own	Delete Any
Owner	✓	✓	✓	✓	✓
Admin	✓	✓	✓	✓	✓
Moderator	✓	✓	✗	✓	✗
Member	✓	✓	✗	✓	✗
Guest	✗	✗	✗	✗	✗

All permissions remain customizable through the Roles & Permissions system.

Core Features

- Rich text editor
- Headings
- Paragraphs
- Bullet lists
- Numbered lists
- Checklists
- Hyperlinks
- @Mentions
- Embedded Media
- Attached Files
- Comments
- Search
- Cross-module linking
- End-to-End Encryption (E2EE)

Cross-Module Linking

Notes can reference:

Current Modules

- Events
- Lists
- Rooms
- Announcements
- Media
- Files
- Polls

Future Modules

- Projects
- Boards
- Bids
- Reports
- Library

Example

Vacation Planning

Linked Event

- Summer Vacation

Linked List

- Packing Checklist

Linked Media

- Cabin Photos

Linked Files

- Rental Agreement.pdf

Notes

Arrival Time:

Check-in begins at 4:00 PM.

Things to Remember:

- Bring fishing gear
- Pick up groceries before arrival
- Cabin Wi-Fi password is provided at check-in.

Everything remains connected without duplicating information.

Final Design Decisions

1. Editing Permissions

Notes follow the standard Roles & Permissions system.

Who can create, edit, or delete Notes depends on the user's role and any individual permission overrides.

There is no separate collaborative editing model outside of the existing permission system.

2. Personal Organization

Notes are organized on a per-user basis.

Each member can organize and view Notes independently without affecting anyone else's experience.

Supported organization methods may include:

- Manual ordering
- Recently viewed
- Most viewed
- Recently updated

This allows Notes to naturally surface the information each member uses most often.

3. Templates

Notes do not support templates in v2.1.

Users create Notes manually.

4. Comments

Notes support comments.

Comments provide a place for discussion without requiring members to edit the Note itself.

Examples:

- Requesting an update
- Reporting outdated information
- Asking a question
- Suggesting additional information

Comments are intended for discussion, while the Note remains the authoritative source of information.

5. Pinning

Notes do not support pinning.

Because each member can organize Notes independently, pinning is unnecessary.

Personal organization replaces the need for pinned Notes.

Spaces v2.5

Module Category 4: Operations

Operations modules are independently permissioned.

Enabling the Operations category does not grant access to every Operations module. Each module has its own permissions, allowing organizations to expose only the workflows appropriate for each role.

Sensitive business information remains isolated to the modules where it belongs.

1. Projects

Purpose

Projects provide a centralized workspace for planning, organizing, tracking, and completing work within a Space.

A Project serves as the primary container for all work related to a specific goal, bringing together tasks, discussions, files, media, reports, and other resources into one connected location.

Projects are intended primarily for Business and Community Spaces but may be enabled in any Space template.

Why?

Projects organize work around a single objective instead of scattering information across multiple modules.

Business

- Website Redesign
- Office Renovation
- Product Launch
- Marketing Campaign

Community

- Annual Fundraiser
- Volunteer Recruitment
- Food Drive
- Park Cleanup

Family

- Home Remodel

- Family Reunion
- Vacation Planning

Permissions

Role	Create	Edit Own	Edit Any	Delete Own	Delete Any
Owner	✓	✓	✓	✓	✓
Admin	✓	✓	✓	✓	✓
Moderator	✓	✓	✗	✓	✗
Member	Optional	Optional	✗	Optional	✗
Guest	✗	✗	✗	✗	✗

All permissions remain fully configurable through the Roles & Permissions system.

Core Features

- Create Projects
- Project description
- Custom project statuses
- Automatic progress tracking
- Project Lead (optional)
- Project member management
- Start date
- Due date
- Cross-module linking
- Search
- End-to-End Encryption (E2EE)

Cross-Module Linking

Projects can reference:

Current Modules

- Rooms
- Notes
- Lists
- Events
- Announcements
- Media
- Files
- Polls

Operations Modules

- Boards
- Bids
- Reports
- Library

Projects serve as the central hub for all work related to a specific objective.

Example

Project

Website Redesign

Project Lead

Ian

Members

- Sarah
- Michael
- Chris

Linked Board

- Website Tasks

Linked Notes

- Brand Guidelines

Linked Files

- Contract.pdf

Linked Media

- Homepage Mockups

Linked Reports

- Weekly Progress Report

Status

In Progress

Progress

72%

Everything associated with the Website Redesign project remains connected through a single Project without duplicating information.

Final Design Decisions

1. Project Statuses

Projects support customizable statuses.

Each Space begins with default statuses, but the Space Owner may:

- Rename statuses
- Reorder statuses
- Add new statuses
- Remove unused statuses
- Customize status colors

This allows every Space to adapt Projects to its own workflow.

Examples:

Default

- Planning
- In Progress
- On Hold
- Completed

Custom

- Backlog
- Sprint
- QA
- Ready for Release
- Complete

2. Progress Tracking

Project progress is calculated automatically.

Progress is determined using information from:

- Board completion
- Current project status
- Future workflow integrations

This ensures project progress always reflects the current state of the work.

3. Project Lead

Projects belong to the Space, not an individual user.

However, a Project may optionally designate a Project Lead responsible for coordinating the work.

The Project Lead does not own the Project—they simply serve as the primary coordinator.

4. Project Members

Each Project maintains its own member list.

Only members assigned to a Project can access and participate in that Project.

Different Projects within the same Space may have completely different member lists.

5. Project Visibility

Project visibility is determined by Project membership.

There are no separate Public or Private Project settings.

If a member belongs to the Project, they can access it.

If they are not a member, the Project is not visible to them.

******add a Project Settings section where things like custom statuses, automations, archived state, notifications, and default board behavior live. That keeps the main Project interface focused on the work itself while advanced configuration stays tucked away in one place. I think that'll scale much better as the Operations suite grows.

2. Boards

Boards

Purpose

Boards provide a visual workflow management system for Projects.

Each Board organizes work into customizable columns where members can create, manage, and move cards as work progresses.

Boards are designed to simplify collaboration by making project progress easy to understand at a glance.

Boards are optional within a Project and should only be created when a visual workflow is needed.

Why?

Boards organize work into manageable stages instead of long task lists.

Business

- Website Development
- Marketing Campaign
- Product Launch
- Hiring Process

Community

- Event Planning
- Volunteer Coordination
- Donation Campaign
- Committee Work

Family

- Home Renovation
- Vacation Planning
- Wedding Planning

Permissions

Role	Create	Edit Own	Edit Any	Delete Own	Delete Any
Owner	✓	✓	✓	✓	✓
Admin	✓	✓	✓	✓	✓
Moderator	✓	✓	✗	✓	✗
Member	Optional	Optional	✗	Optional	✗
Guest	✗	✗	✗	✗	✗

All permissions remain fully configurable through the Roles & Permissions system.

Core Features

- Create Boards
- Multiple Boards per Project
- Customizable workflow columns
- Create Cards
- Drag and drop cards between columns
- Press and hold to move cards on mobile

- Rich text card editor
- Multiple assignees
- Labels
- Priority
- Due dates
- Comments
- Activity history
- Cross-module linking
- Search
- End-to-End Encryption (E2EE)

Card Structure

Each Board Card consists of four sections.

Metadata

- Title
- Status
- Assignee(s)
- Priority
- Labels
- Due Date
- Parent Project

Content

The card body uses the same rich text editing experience used throughout Spaces.

Supported content includes:

- Headings
- Paragraphs
- Bullet lists
- Numbered lists
- Checklists
- Tables
- Quotes
- Code blocks
- Hyperlinks
- @Mentions
- Embedded Media
- Attached Files
- Cross-module links

Comments

Cards include a dedicated discussion section.

Comments allow members to collaborate, ask questions, and discuss work without modifying the card's content.

Activity

Each card maintains an automatic activity history.

Examples:

- Card created
- Card edited
- Member assigned
- Due date changed
- Card moved
- Files uploaded
- Status changed

Activity history is generated automatically and cannot be edited.

Cross-Module Linking

Board Cards can reference:

Current Modules

- Projects (Parent)
- Notes
- Lists
- Events
- Rooms
- Announcements
- Media
- Files
- Polls

Future Modules

- Reports
- Bids
- Library

Board Cards reference information rather than duplicating it, following the Module Interoperability Principle.

Example

Project

Website Redesign

Board

Development

Columns

Backlog

- Homepage Design
- Contact Page

In Progress

- Mobile Navigation

Review

- Pricing Page

Completed

- Footer Redesign

Selecting a card opens its full details, including metadata, rich content, comments, linked resources, and activity history.

Final Design Decisions

1. Boards

Boards are optional.

Projects may contain zero, one, or multiple Boards depending on the project's needs.

2. Workflow Columns

Boards support fully customizable columns.

Users with appropriate permissions may:

- Create columns
- Rename columns
- Reorder columns
- Delete columns

Each Board defines its own workflow independently.

3. Card Movement

A card may exist in only one column at a time.

Cards move through the workflow by:

- Drag and drop
- Press and hold (mobile)

Moving a card automatically updates its workflow status and records the action in the card's activity history.

4. Rich Content

Board Cards are not limited to short descriptions.

Each card contains a rich document that supports the same editing capabilities used throughout Spaces.

This allows cards to contain detailed documentation while remaining part of a visual workflow.

5. Card Assignments

Cards may be assigned to one or more members.

Assignments use @mentions and are displayed within the card metadata.

6. Due Dates

Boards may optionally define an overall due date.

Individual cards may also define their own due dates.

This allows both project-level deadlines and task-level deadlines to coexist.

7. Simplicity

Boards are designed to provide visual workflow management.

Advanced enterprise project management features such as task dependencies, blockers, workflow automations, and Gantt charts are intentionally outside the scope of v2.5.

3. Bids

Purpose

Bids provide a centralized workflow for managing proposals, estimates, approvals, invoices, and client negotiations before work begins.

Bids allow organizations to move opportunities through a customizable workflow and seamlessly convert approved work into Projects without duplicating information.

Bids are intended primarily for Business Spaces and may be enabled in any Space template.

Why?

Bids organize the entire proposal process into a single workflow instead of separating proposals, approvals, invoices, and estimates into multiple modules.

Business

- Website Proposal
- Marketing Campaign
- Product Development
- Consulting Services

Community

- Sponsorship Requests
- Grant Proposals
- Vendor Quotes

Permissions

Role	Create	Edit Own	Edit Any	Delete Own	Delete Any
-----	:-----	:-----	:-----	:-----	:-----
Owner					
Admin					
Moderator	Optional	Optional		Optional	
Member					
Guest					

All permissions remain fully configurable through the Roles & Permissions system.

Core Features

- Create Bids

- Custom workflow columns
- Proposal management
- Estimates and pricing
- Client information
- Approval tracking
- Invoice management
- Rich text editor
- Comments
- Activity history
- Convert Bid to Project
- Cross-module linking
- Search
- End-to-End Encryption (E2EE)

Bid Card Structure

Each Bid Card consists of four sections.

Metadata

- Title
- Client
- Status
- Proposal Value
- Expiration Date
- Assigned Member(s)
- Approval Status

Content

The Bid uses the same rich text editing experience used throughout Spaces.

Supported content includes:

- Headings
- Paragraphs
- Bullet lists
- Numbered lists
- Checklists
- Tables
- Quotes
- Code blocks
- Hyperlinks
- @Mentions
- Embedded Media
- Attached Files
- Cross-module links

This section typically contains:

- Proposal
- Scope of Work
- Timeline
- Deliverables
- Pricing
- Terms and Conditions

Comments

Bid Cards include a dedicated discussion section.

Comments allow internal collaboration without modifying the proposal itself.

Activity

Each Bid maintains an automatic activity history.

Examples:

- Bid created
- Proposal edited
- Status changed
- Client updated
- Approval recorded
- Invoice added
- Converted to Project

Activity history is generated automatically and cannot be edited.

Invoice Management

Invoices live within the Bid.

A Bid may contain one or more invoices, allowing organizations to manage deposits, milestone payments, and final payments without requiring a separate module.

Invoices remain associated with the originating Bid throughout its lifecycle.

Approval Management

Bids support approval tracking.

Approval statuses include:

- Pending
- Approved

- Declined

Approvals may include:

- Approval date
- Approved by
- Notes

This supports both internal and external approval workflows.

Cross-Module Linking

Bid Cards can reference:

Current Modules

- Notes
- Lists
- Events
- Rooms
- Announcements
- Media
- Files
- Polls

Operations Modules

- Projects
- Boards
- Reports
- Library

Bid Cards reference information rather than duplicating it, following the Module Interoperability Principle.

Example

Business Development

Columns

Draft

- Website Proposal

Sent

- Mobile App Proposal

Review

- Marketing Campaign

Approved

- Restaurant Website

Converted

- Bakery Website

Archived

- City Hall Redesign

Selecting a Bid opens its metadata, proposal, invoices, approvals, comments, linked resources, and activity history.

Final Design Decisions

1. Workflow

Bids use customizable workflow columns.

Organizations may:

- Create columns
- Rename columns
- Reorder columns
- Delete columns

Each organization defines its own sales workflow.

2. Bid Card Experience

Bid Cards use the same interaction model as Board Cards.

Each Bid contains:

- Metadata
- Rich Content
- Comments
- Activity

This provides a consistent experience across Operations modules while allowing each module to define its own metadata.

3. Convert to Project

Approved Bids may be converted into Projects.

Users with appropriate permissions may:

- Create a new Project
- Link the Bid to an existing Project

The Bid remains in the Bids module as the permanent historical record.

The new Project contains a reference back to the originating Bid.

No information is duplicated.

4. Invoice Management

Invoices are managed within the Bid.

A Bid may contain multiple invoices throughout its lifecycle.

Invoices are considered supporting records rather than independent workflow items.

5. Approval Tracking

Approvals are tracked within each Bid.

Approval records remain permanently attached to the Bid for historical reference and auditing.

6. Historical Record

Converting a Bid into a Project does not remove the Bid.

The Bid remains available for future reference, including:

- Proposal
- Pricing
- Client approvals
- Invoices
- Activity history
- Linked Project

This preserves the complete business history of every Project from initial proposal through completion.

4. Reports

Purpose

Reports provide structured, formal records that document specific operational events, findings, incidents, and investigations.

Unlike Library articles, Reports capture what happened at a specific point in time. Once finalized, they become part of the organization's historical record.

Reports are primarily organization-focused rather than client- or project-focused, but may reference Projects, Boards, Bids, and other modules when appropriate.

Why?

Reports help organizations document operational events, maintain accountability, preserve evidence, and create permanent historical records.

Examples include:

- Incident Reports
- Accident Reports
- Damage Reports
- Investigation Reports
- Audit Reports
- Compliance Findings
- Security Reports

Permissions

Role	Create	Edit Draft	Finalize	View	Comment	Archive	Delete
Owner	✓	✓	✓	✓	✓	✓	✓
Admin	✓	✓	✓	✓	✓	✓	✓
Moderator	Optional	Optional	Optional	Optional	Optional	Optional	✗
Member	Optional	Optional	✗	Optional	Optional	✗	✗
Guest	✗	✗	✗	Optional	✗	✗	✗

All permissions remain fully configurable through the Roles & Permissions system.

Core Features

- Create Reports
- Rich text editor

- Organization-defined templates
- Draft, Finalized, and Archived workflow
- Comments
- Activity history
- Cross-module linking
- Search
- End-to-End Encryption (E2EE)

Report Structure

Each Report consists of four sections.

Metadata

- Title
- Report Status
- Created By
- Created Date
- Last Updated
- Finalized Date (optional)
- Tags
- Organization-defined fields

Content

Reports use the same rich text editing experience used throughout Spaces.

Supported content includes:

- Headings
- Paragraphs
- Bullet lists
- Numbered lists
- Checklists
- Tables
- Quotes
- Code blocks
- Hyperlinks
- @Mentions
- Embedded Media
- Attached Files
- Cross-module links

Comments

Reports support comments for collaboration and review.

Commenting is permission-based and controlled through the Roles & Permissions system.

Comments allow discussion without altering the report itself.

Activity

Each Report maintains an automatic activity history.

Examples:

- Report created
- Report edited
- Report finalized
- Report archived
- Files attached
- Linked resources added
- Comments added

Activity history is generated automatically and cannot be edited.

Report Workflow

Reports progress through a simple lifecycle.

Draft

↓

Finalized

↓

Archived

Draft

Reports remain editable while in Draft status.

Finalized

Finalized Reports become official organizational records.

They are locked from further editing unless reopened by users with appropriate permissions.

Archived

Archived Reports remain searchable and available for historical reference.

Organization Templates

Reports support organization-defined templates.

Organizations may create templates that match their own operational reporting needs.

Examples:

ArcInteractive

- Employee Incident
- Security Incident
- Investigation
- Damage Report

Utility Company

- Storm Damage Report
- Equipment Failure
- Outage Investigation
- Compliance Finding

Church

- Volunteer Incident
- Property Damage
- Safety Incident

Spaces does not provide predefined report templates.

Organizations define the templates that best fit their operations.

Cross-Module Linking

Reports can reference:

Current Modules

- Notes
- Lists
- Events
- Rooms
- Announcements
- Media
- Files
- Polls

Operations Modules

- Projects
- Boards
- Bids
- Library

Reports reference information rather than duplicating it, following the Module Interoperability Principle.

Example

Operations

Reports

- Employee Incident
- Security Investigation
- Storm Damage Report
- Annual Compliance Audit
- Property Damage Report

Selecting a Report opens its metadata, report content, linked resources, comments, and activity history.

Final Design Decisions

1. Purpose

Reports document operational events, findings, incidents, investigations, and other point-in-time records.

They are not intended for ongoing documentation or collaborative knowledge management.

2. Organization Focus

Reports belong to the organization itself.

They may reference Projects, Boards, Bids, or other modules when appropriate but are not owned by those modules.

3. Workflow

Reports use a simple lifecycle:

- Draft
- Finalized

- Archived

This preserves the integrity of official organizational records.

4. Organization Templates

Organizations may create and manage their own report templates.

Spaces does not include predefined templates.

5. Rich Content

Reports support the same rich text editing capabilities used throughout Spaces, allowing organizations to create detailed operational records.

6. Comments

Reports support permission-based comments for review and collaboration without modifying the report itself.

7. Historical Records

Reports remain within the Reports module for their entire lifecycle.

They do not become Library articles.

Reports document what happened.

Library articles document how things are done.

5. Library

Purpose

Library provides a centralized knowledge base for an organization.

Unlike Notes, which are collaborative working documents, and Reports, which document specific events, Library stores long-term organizational knowledge that evolves over time.

Library is designed as a wiki-style knowledge system where information is organized into nested pages and remains easily searchable and accessible.

Why?

Library preserves institutional knowledge and provides a single place for members to learn how the organization operates.

Examples include:

- Employee Handbook
- Company Policies
- Standard Operating Procedures (SOPs)
- Training Guides
- Onboarding Documentation
- IT Documentation
- Brand Guidelines
- Best Practices
- Process Documentation
- Frequently Asked Questions (FAQs)

Permissions

Role	Create	Edit Draft	Publish	View	Comment	Archive	Delete
Owner	✓	✓	✓	✓	✓	✓	✓
Admin	✓	✓	✓	✓	✓	✓	✓
Moderator	Optional	Optional	Optional	Optional	Optional	Optional	✗
Member	Optional	Optional	✗	Optional	Optional	✗	✗
Guest	✗	✗	✗	Optional	✗	✗	✗

All permissions remain fully configurable through the Roles & Permissions system.

Core Features

- Wiki-style knowledge base
- Nested pages
- Rich text editor
- Internal page linking
- Version history
- Draft, Published, and Archived workflow
- Comments
- Activity history
- Cross-module linking
- Search
- End-to-End Encryption (E2EE)

Library Article Structure

Each Library article consists of four sections.

Metadata

- Title
- Status
- Created By
- Created Date
- Last Updated
- Published Date (optional)
- Tags

Content

Library articles use the same rich text editing experience used throughout Spaces.

Supported content includes:

- Headings
- Paragraphs
- Bullet lists
- Numbered lists
- Checklists
- Tables
- Quotes
- Code blocks
- Hyperlinks
- @Mentions
- Embedded Media
- Attached Files
- Cross-module links

Comments

Library articles support comments for collaboration and review.

Commenting is permission-based and controlled through the Roles & Permissions system.

Activity

Each Library article maintains an automatic activity history.

Examples:

- Article created
- Article edited
- Published

- Archived
- Version restored
- Linked resources added
- Comments added

Activity history is generated automatically and cannot be edited.

Wiki Organization

Library uses a wiki-style structure.

Knowledge is organized into nested pages rather than folders.

Top-level articles act as the entry points for related documentation.

Example

Employee Handbook

- PTO
- Benefits
- Holidays
- Code of Conduct

Operations Manual

- Opening Procedures
- Closing Procedures
- Inventory
- Safety

IT Documentation

- Email Setup
- VPN
- Password Policy

Pages may contain child pages up to a maximum depth of three levels.

Workflow

Library articles progress through a simple lifecycle.

Draft

↓

Published



Archived

Draft

Articles remain editable while in Draft status.

Published

Published articles become the organization's official knowledge source.

They remain editable by users with appropriate permissions.

Each edit automatically creates a new version.

Archived

Archived articles remain searchable and available for historical reference.

Version History

Every published change creates a new version.

Version history is accessible from within each Library article.

Previous versions open in a read-only document view.

Users with appropriate permissions may restore an earlier version.

The complete version history remains preserved.

Internal Linking

Library articles may link directly to other Library articles.

Examples:

See Also

- Password Policy
- VPN Setup
- Employee Handbook

This allows organizations to build an interconnected knowledge base.

Cross-Module Linking

Library articles can reference:

Current Modules

- Notes
- Lists
- Events
- Rooms
- Announcements
- Media
- Files
- Polls

Operations Modules

- Projects
- Boards
- Bids
- Reports

Library articles reference information rather than duplicating it, following the Module Interoperability Principle.

Example

Library

Employee Handbook

- PTO
- Benefits
- Holidays
- Conduct

Operations Manual

- Opening Procedures
- Closing Procedures
- Safety Procedures

IT Documentation

- Email Setup
- VPN
- Password Policy

Selecting an article opens its content, linked resources, version history, comments, and activity history.

Final Design Decisions

1. Purpose

Library stores long-term organizational knowledge.

It documents how the organization operates rather than recording specific events.

2. Wiki Structure

Library uses nested pages instead of folders.

Top-level articles serve as the root for related documentation.

Page nesting is limited to a maximum of three levels to keep navigation manageable.

3. Workflow

Library articles use a simple lifecycle:

- Draft
- Published
- Archived

Published articles represent the organization's official documentation.

4. Version History

Every published edit creates a new version.

Previous versions remain available, can be viewed in a read-only document view, and may be restored by users with appropriate permissions.

5. Internal Linking

Library articles support links to other Library articles, creating an interconnected knowledge base.

6. Rich Content

Library supports the same rich text editing capabilities used throughout Spaces, allowing organizations to create comprehensive documentation.

7. Living Knowledge

Unlike Reports, Library articles are intended to evolve over time.

Library documents how the organization operates.

Reports document what happened.